TECHNICAL FOLLOW-UP PACKAGE | JULY 16, 2026

LumenCore + LANL VISION

A bounded licensing, validation, and commercialization fit for passive industrial vibration intelligence with reproducible proof-to-pilot evidence.

Decision requested: Is there a credible path to a short technical diligence session and an evaluation-stage relationship for VISION, with the appropriate LANL licensing, NDA, data, and pilot boundaries established first?

THE FIT

VISION supplies independent physical evidence

LANL publicly describes VISION as a passive, control-system-independent vibration monitoring technology for pumps, compressors, fans, and related industrial equipment.

THE LAYER

LumenCore supplies the audit and pilot spine

LumenCore can organize source identity, baseline definitions, locked metrics, shadow-mode evaluations, adverse-result retention, hashes, and reviewer-facing result packets around a jointly defined VISION evaluation.

Why the combination is commercially useful

- Physical vibration data can provide an independent check on networked control-system telemetry.
- A proof spine can separate data provenance, model behavior, operational decisions, and claims.
- Shadow-mode deployment supports evaluation before any control or maintenance action is authorized.
- Locked baselines and holdouts make pilot acceptance criteria explicit before scoring.
- Null, adverse, and error rows remain in the evidence record rather than being discarded.
- The same evidence structure can support future licensee, pilot-customer, grant, or partner diligence.

CLAIM BOUNDARY

LumenCore does not claim ownership of VISION, a LANL license, LANL endorsement, field validation, cybersecurity certification, realized savings, or production readiness. This package proposes a diligence and evaluation path only.

Prepared for Michael Erickson and Neil Loychik, Los Alamos National Laboratory
Prepared by Robert Ashworth, Founder, LumenCore | robertashworth4444@gmail.com | lumen-core.ai

COMPLEMENTARY SYSTEM ARCHITECTURE

Keep sensing, evidence, and action authority separate

The proposed architecture treats VISION as the physical-sensing and machine-state signal layer. LumenCore does not replace VISION's sensing logic or the facility's control system; it adds a governed evaluation and evidence path.

LANL VISION

- Machine-mounted vibration sensing
- Machine-specific signature learning
- Performance deviation indicators
- Independent physical feedback
- Interfaces and data rights defined by LANL

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LumenCore evidence layer

- Source and schema registration
- Timestamp and integrity checks
- Named baseline and candidate registry
- Predeclared metrics and holdout windows
- Shadow-mode evaluation
- Hash-sealed result and exception records

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Reviewer / operator

- Inspectable pilot packet
- Accepted or rejected gate outcome
- Traceable deviations and limitations
- Human maintenance authority
- No automatic actuation by default

Default operating mode

Read-only observation and replay. Any maintenance recommendation, alert routing, integration, or control action remains separately authorized by the responsible owner.

Evidence object

Each result binds source identity, baseline, candidate, metric, time window, protocol version, environment receipt, exceptions, and terminal hash.

Example initial use cases

Use case	Candidate evidence question	Safe first mode
Pumps and cooling systems	Does VISION detect meaningful deviation earlier or more reliably than an accepted incumbent signal?	Historical replay, then shadow monitoring
Data-center mechanical infrastructure	Can the independent vibration channel improve confidence when network telemetry is incomplete or suspect?	Read-only parallel observation
Industrial cybersecurity	Can physical feedback identify inconsistencies under a jointly defined benign fault or spoofing test?	Controlled lab scenario only

PROPOSED EVALUATION STRUCTURE

A staged 90-day proof-to-pilot path

Timing is illustrative and would begin only after LANL identifies the appropriate agreement route, disclosure boundary, and technical owner.

Stage	Purpose	Work product	Exit gate
0 Weeks 0-2	Licensing and disclosure fit	Evaluation route, NDA/export review, field-of-use discussion, interface inventory, and named technical contacts.	Written authorization for the bounded evaluation scope.
1 Weeks 2-4	Protocol lock	Accepted dataset or simulator, incumbent baseline, metric definitions, holdout window, exclusion rules, and failure criteria.	Protocol hash and written acceptance before scoring.
2 Weeks 4-8	Retrospective or laboratory evaluation	Read-only adapter, quality report, baseline comparison, exception ledger, and complete null/adverse result retention.	Reproducible output package with no operational claim.
3 Weeks 8-12	Independent review and next decision	Blind reproduction or reviewer replay, signed deviations, terminal hashes, and a pilot recommendation.	Stop, revise, or approve a separate shadow-site pilot.

Acceptance packet

01
Protocol and source manifest

02
Baseline and metric register

03
Raw and exception-preserving outputs

04
Hashes and environment receipt

What would count as progress

- LANL accepts the evaluation question and baseline.
- The bounded test can be independently rerun.
- The result and its limitations survive technical review.
- A specific license or pilot next step becomes decision-ready.

What would not count as progress

- Internal reruns of the same evidence presented as independent validation.
- Changing metrics or holdouts after seeing results.
- Converting diagnostic outputs into savings or safety claims.
- Using proprietary VISION information outside its authorized purpose.

DILIGENCE AGENDA

Questions for the next technical discussion

1. **Licensing route:** Is VISION available for evaluation, option, field-of-use, or commercial licensing, and what information does LANL need from a prospective small-business partner?
2. **Technical boundary:** Which sensing hardware, firmware, models, calibration methods, interfaces, and documentation are included or excluded?
3. **Data path:** Is there a representative, releasable dataset or controlled test environment suitable for a first baseline-locked evaluation?
4. **Acceptance standard:** Which incumbent sensor or operational signal, metrics, operating conditions, and failure rules would LANL regard as a credible comparison?
5. **Protection and compliance:** Should deeper discussion begin under an NDA, and what export-control, cybersecurity, publication, or background-IP constraints must be screened first?
6. **Commercialization path:** Would LANL prefer a licensing-only discussion, an evaluation agreement, an SPP/CRADA route, a pilot customer, or a commercialization-partner process?

LUMENCORE CAN CONTRIBUTE

- Reviewer-safe pilot design
- Data and schema adapters
- Baseline and holdout locking
- Shadow-mode evaluation
- Manifest, hash, and exception records
- Grant and partner diligence packaging

NOT REPRESENTED

- Existing rights to VISION
- Hardware manufacturing readiness
- Field or customer validation
- Cybersecurity or safety certification
- Guaranteed performance or savings
- Patent coverage or freedom to operate

Requested next action

A 30-minute follow-up with the appropriate licensing and technical participants to decide whether Stage 0 is warranted, followed by a bounded protocol in LANL's preferred format.

References and reviewer links

LANL public VISION description: National Security Science, Summer 2025
LumenCore public proof gateway: https://lumen-core.ai/proof_to_pilot.html
LumenCore public repository: https://github.com/robertashworth1986-debug/lumen-core-public

Contact

Robert Ashworth | Founder, LumenCore
robertashworth4444@gmail.com | https://lumen-core.ai